SecureRandom is one of the **most important security classes in Java**. It is widely used in **banking applications, Spring Security, JWT authentication, RSA key generation, OTP generation, session IDs, and cryptography**.

What is SecureRandom?

SecureRandom is a Java class that generates **cryptographically strong random numbers**.

Package:

```
import java.security.SecureRandom;
```

Unlike Random, SecureRandom produces numbers that are extremely difficult to predict.

Why Do We Need SecureRandom?

Imagine you're generating a 6-digit OTP.

If you use a predictable random number generator, an attacker may be able to guess future OTPs.

Example:

```
Random random = new Random();
```

This is **not recommended for security** because Random uses a deterministic algorithm (a seed-based pseudo-random generator).

For security-sensitive values like:

- OTPs
- Password reset tokens
- Session IDs
- JWT secrets
- RSA keys
- AES keys

you should use:

```
SecureRandom secureRandom = new SecureRandom();
```

Example 1 – Generate a Random Number

```
import java.security.SecureRandom;
public class Demo {
  public static void main(String[] args) {
    SecureRandom secureRandom = new SecureRandom();
    int number = secureRandom.nextInt(100);
    System.out.println(number);
  }
}
```

The result values are designed to be difficult to predict.

Difference Between Random and SecureRandom

Random	SecureRandom
Package: java.util.Random	Package: java.security.SecureRandom
Fast	Slightly slower
Predictable if the seed is known	Designed to be unpredictable
Good for games and simulations	Good for security and cryptography
Not suitable for passwords or OTPs	Suitable for passwords, OTPs, tokens, keys

Where Is SecureRandom Used?

01) OTP Generation
02) Session ID Generation
03) JWT Secret Generation
04) RSA Key Pair Generation
05)AES Key Generation

Why Not Use Random?

Suppose someone writes:

```
Random random = new Random(123);
```

Every time the program runs, it generates the same sequence.

If an attacker knows the seed, they can predict future values.

SecureRandom in Banking

In banking applications, SecureRandom is commonly used to generate:

- Transaction reference numbers (when randomness is required)
- OTPs
- Password reset links
- Authentication tokens
- Session IDs
- Encryption keys
- Nonces used in cryptographic protocols

Interview Answer

What is SecureRandom in Java?

SecureRandom is a class in the java.security package that generates cryptographically strong random numbers. It is designed for security-sensitive applications such as OTP generation, session IDs, authentication tokens, encryption keys, and digital signatures. Unlike java.util.Random, it produces values that are intended to be unpredictable, making it suitable for use in authentication and cryptographic operations.